



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam: Spring 2026

Course Code: CSE 3811, Course Title: Artificial Intelligence

Total Marks: 30

Duration: 1 hour 30 minutes

Answer all questions. Marks are indicated on the right side of each question.

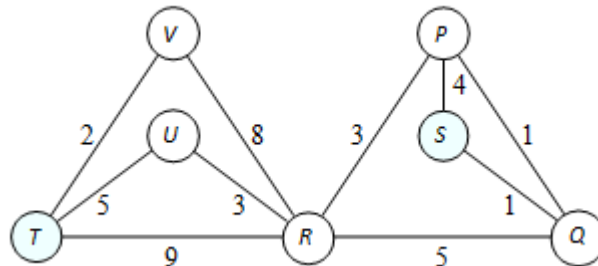
[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

1. A robot arm sits at a postal sorting station. Letters arrive one at a time on a tray. Each letter has a destination district (A, B, C, or D) printed on it, and the robot must pick up the letter and drop it into one of four bins. The robot's OCR scanner misreads the district label 15% of the time, randomly confusing one letter for another (e.g., reading "B" as "D"). The robot can move between 5 fixed positions: the tray and the four bin locations. Moving between any two adjacent positions costs 1 second. The robot earns +5 for a correct bin and -10 for a wrong bin.

[2]

 - a. Why is the environment of the above scenario **Episodic** and **Stochastic**? Give proper reasoning for both in context of the scenario.
 - b. Formulate the above scenario as a search problem.

[4]
2. Consider the following state space graph

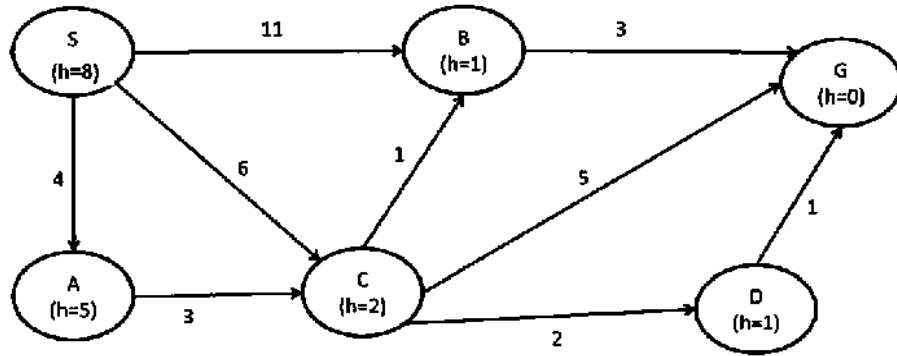


In the given figure above, *S* is the start state and *T* is the goal state. The cost for taking each edge (in both directions) is shown on the graph. We can apply *Graph Search Strategy* to find path from start state to the goal state using:

- i. Depth First Search
 - ii. Breadth First Search
 - iii. Uniform Cost Search
- [5]
- a. Which of the following paths will be returned for each of the algorithms mentioned above? Understand that that for some of the algorithms mentioned above, the path from the start state to the goal state can vary based on how we break ties among multiple options. In such cases, mention *all* possible paths returned.
 1. $S \rightarrow P \rightarrow R \rightarrow T$
 2. $S \rightarrow Q \rightarrow R \rightarrow T$
 3. $S \rightarrow Q \rightarrow P \rightarrow R \rightarrow U \rightarrow T$
 - b. Which algorithm(s) will find the least cost path?

[1]

3. Consider the following directed Search Space, where S is the Initial State and G is the Goal State:



- Simulate the following algorithms and show the solution path and cost (In case of ties, consider alphabetical order):
 - A* Tree Search
 - Greedy Best First Search
- If we apply A* graph search, do you think you will get the optimal solution for the given graph? What's the reason? [2]

4. The following figure (4.1) depicts a **local search problem (maximization)** and Garfield the lazy cat is using **simulated annealing** to find the best cat food. Each node has a utility value/Energy of cat food associated with it. Temperature is **T = 2 Unit**. And Random Number is generated (RNG) using the formula:

$$RNG = |\sin(25 + 5t)| \text{ [sine is in degrees and } t=\text{timestep}]$$

A is the current node at timestep $t = 0$. For each node checking, the timestep increases by 1. And Nodes are evaluated in alphabetical order; therefore, at timestep $t = 1$ node B is checked.

Figure 4.1

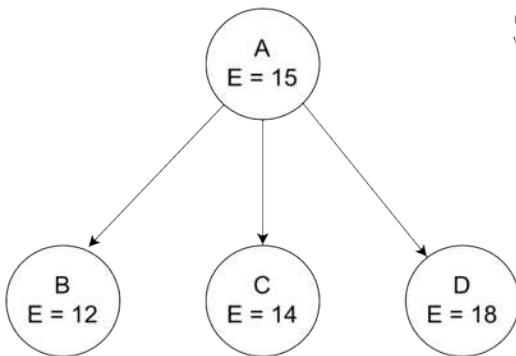
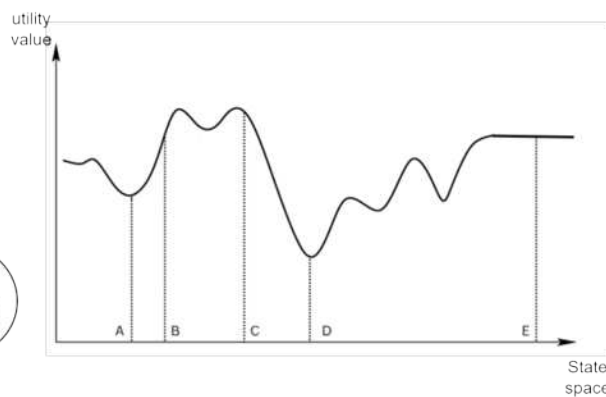


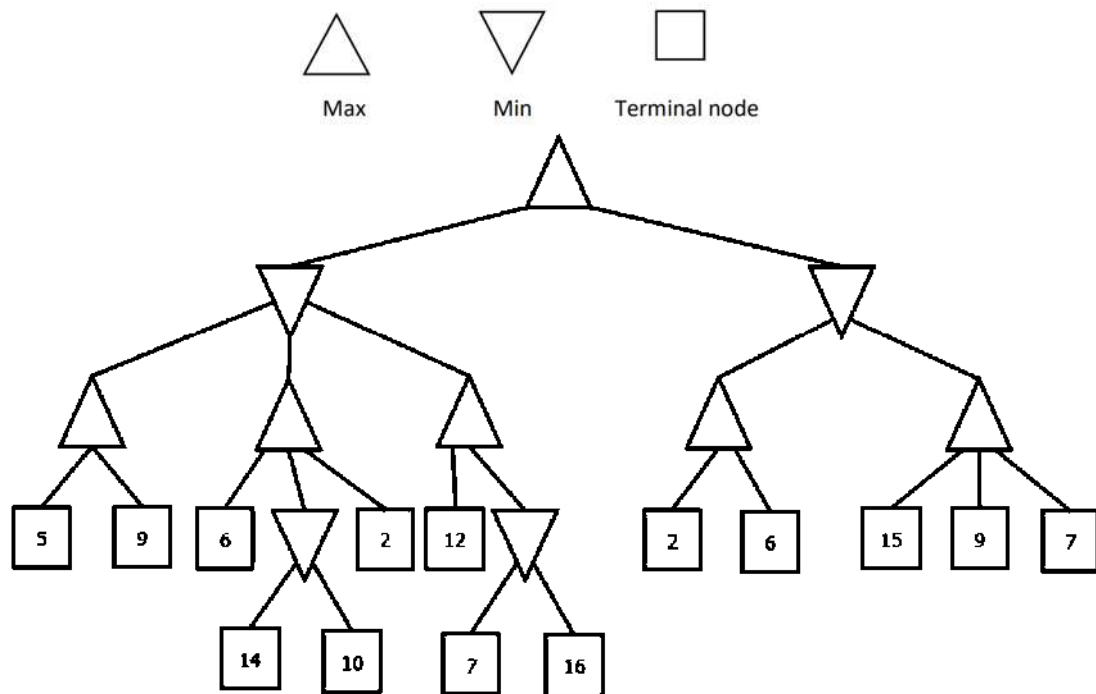
Figure 4.2



- Determine the **first neighbor node** of A that will be **accepted/selected** (Figure 4.1) [2]
- Find the **minimum Temperature** at which node **B** will be **selected**. (Figure 4.1) [2]
- In Figure 4.2, a **basic Hill Climbing algorithm** is applied to a minimization [2]

problem. Using node annotations, provide **any two scenarios** where the algorithm may encounter issues.

5.



- a. For the given game tree, show which nodes will be pruned if you use minimax search algorithm with alpha-beta pruning. Clearly show the values of each node. [5]
- b. Is it possible to have **no pruning at all** in Alpha-Beta? When does this happen? [1]